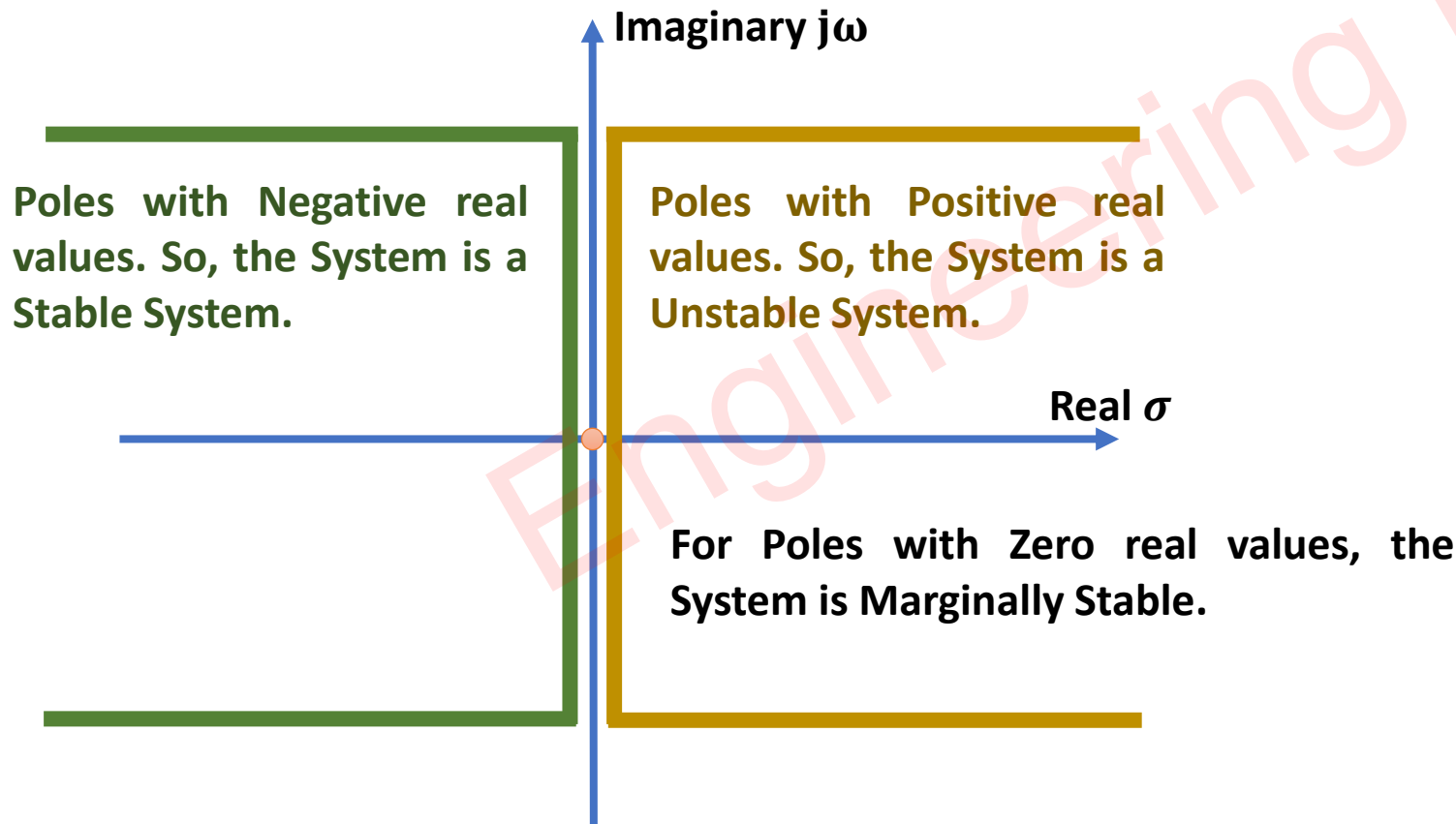


Rules of Root Locus

Basics of Root Locus

- ❑ It plots the System's Dynamic Characteristics.
- ❑ It explains the plot of systems roots (Poles and Zeros) on S-Plane.
- ❑ It explains the system's response and region of stability on S-Plane.



Rules of Root Locus

- ❑ Root Locus Plot emerges from the poles and gets enclosed to the Zeros or Asymptotes.
- ❑ Root Locus must be symmetric with respect to the real Axis.
- ❑ All the Steps are not Compulsory, but based on requirements we need to calculate steps.
- ❑ Step – 1: Identify the Roots (Poles and Zeros) of the System. Total Loci = Max (P, Z)
- ❑ Step – 2: Number of Asymptotes $X = \text{Total Poles } P - \text{Total Zeros } Z$
- ❑ Step – 3: Angle of Asymptotes = $\frac{(2n+1)}{P-Z} 180^\circ$, Where, $n = 0, 1, 2, \dots, (X-1)$
- ❑ Step – 4: Centroid of Asymptotes $\sigma_c = \frac{\sum \text{Real Part of Poles} - \sum \text{Real Part of Zeros}}{P-Z}$
- ❑ Step – 5: Find Break Away Points.
 - Find Characteristics equation $[1 + G(S)H(S) = 0]$
 - Compute $K = \text{Polynomial}$
 - Compute $\frac{dK}{dS} = 0$ and find $S = \text{Break Away Points}$
- ❑ Step – 6: Find Angle of Departure $\theta_d = 180^\circ - [\sum \theta_p - \sum \theta_z]$

Rules of Root Locus

□ Step – 7: Intersection to Imaginary Axis.

- Find Characteristics equation [$1 + G(S)H(S) = 0$]
- Construct Routh matrix.
- Find K for Marginal stable System. {From 1st Column Routh Matrix}
- Place K in Auxiliary Equation {2nd Order Equation in Routh Matrix}.

Engineering Funda

1. Obtain Root-Locus Plot for the Unity Feedback System with System Gain:

$$\Rightarrow G(S) = \frac{K}{S(S+2)}$$

2) Step-1 - Obtain Loci.

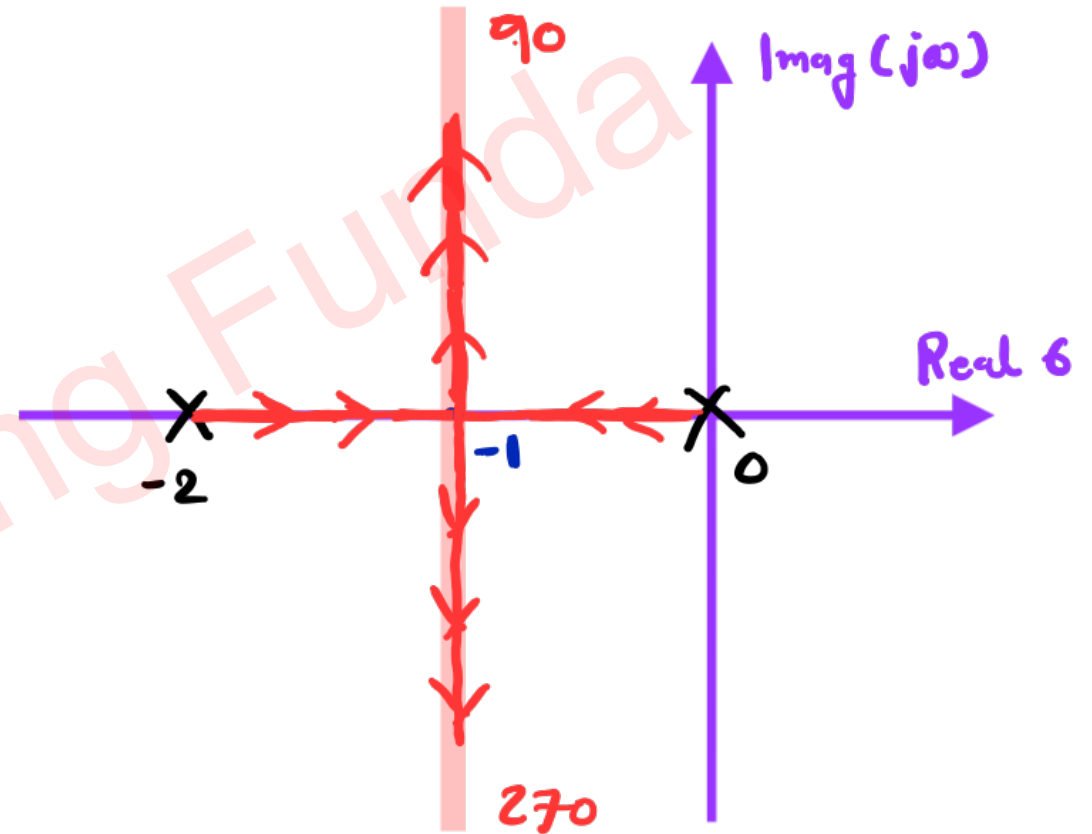
- $P_1 = 0$, $P_2 = -2$
- Total poles $P = 2$
Total zeros $Z = 0$
- Number of Loci = $\text{Max}(P, Z) = 2$

3) Step-2 - Number of Asymptotes.

$$X = P - Z = 2$$

4) Step-3 - Angle of Asymptotes.

$$\theta_a = \frac{(2n+1)}{P-Z} 180^\circ, \quad n = 0, 1 = 90^\circ, 270^\circ$$



~) Step 4 - Centroid of Asymptotes.

$$- \sigma_c = \frac{\sum \text{Real } p - \sum \text{Real } z}{P - Z}$$

$$= \frac{(-2) - 0}{2} = -1$$

~) Step 5 - Break Away point

- Characteristic eq.ⁿ

$$\Rightarrow 1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{K}{s(s+2)} = 0$$

$$\Rightarrow K = -s(s+2)$$

$$\Rightarrow K = -s^2 - 2s$$

$$- \frac{dK}{ds} = 0$$

$$\Rightarrow -(2s) - 2 = 0$$

$$\Rightarrow \boxed{s = -1}$$

2. Obtain Root-Locus Plot for the Unity Feedback System with System Gain:

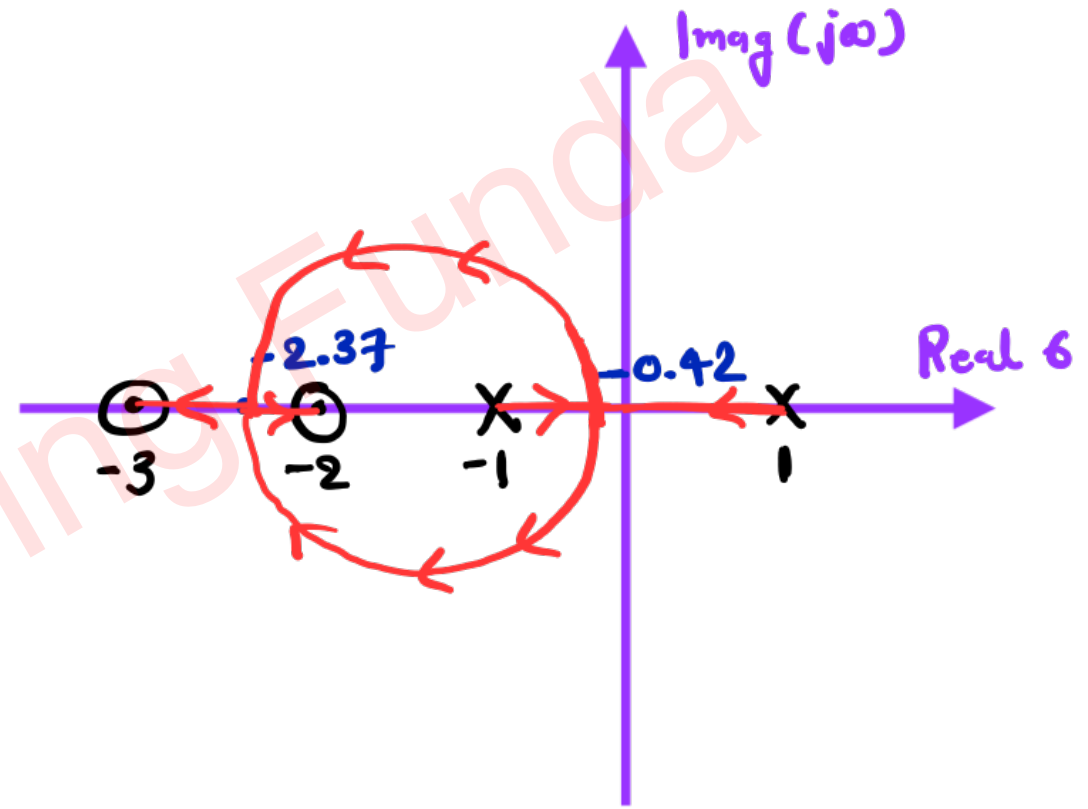
$$\Rightarrow G(S) = \frac{K(S+2)(S+3)}{(S+1)(S-1)}$$

⇒ Step 1 - Obtain Loci

- $z_1 = -2, z_2 = -3$
- $p_1 = -1, p_2 = 1$
- Total Poles $P = 2$
Total Zeros $Z = 2$
- Total loci = $\max(P, Z)$
 $= 2$

⇒ Step 2 - Number of Asymptotes

$$X = P - Z = 2 - 2 = 0$$



~) Step 3 - Break Away points.

~) Characteristics eq.ⁿ

$$\Rightarrow 1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{K(s+2)(s+3)}{(s+1)(s-1)} = 0$$

$$\Rightarrow K = - \frac{(s^2-1)}{(s^2+5s+6)}$$

$$\sim \frac{dK}{ds} = 0 \Rightarrow \frac{(s^2+5s+6)(2s) - (s^2-1)(2s+5)}{(s^2+5s+6)^2} = 0$$

$$\Rightarrow \cancel{2s^3} + 10s^2 + 12s - \cancel{2s^3} - 5s^2 + 2s + 5 = 0$$

$$\Rightarrow 5s^2 + 14s + 5 = 0$$

$$\Rightarrow s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-14 \pm \sqrt{14^2 - 4(5)(5)}}{2 \times 5}$$

$$= -0.42, -2.37$$

3. Obtain Root-Locus Plot for the Unity Feedback System with System Gain:

$$\Rightarrow G(S) = \frac{K(S + 4/3)}{S^2(S + 12)}$$

⇒ Step-1 - Obtain loci

- $z_1 = -4/3$

- $p_1, p_2 = 0, p_3 = -12$

- Total poles $P = 3$

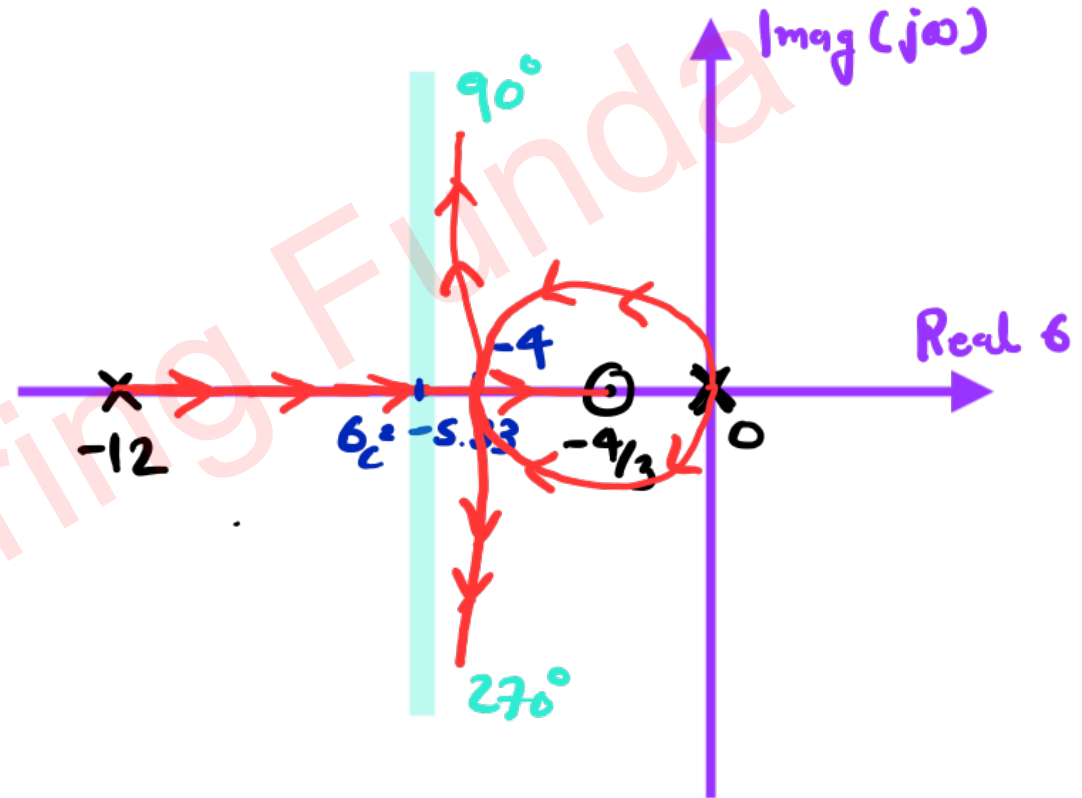
Total Zeros $Z = 1$

- Total loci = $\max(P, Z)$
 $= 3$

⇒ Step-2 - Total Asymptotes.

- $X = P - Z = 3 - 1 = 2$

⇒ Step-3 - Angle of Asymptotes.



$$- \theta = \frac{(2n+1)}{P-2} 180^\circ, \quad n = 0, 1$$

$$= 90^\circ, 270^\circ$$

⇒ Step 4 - Centroid of Asymptotes.

$$- \sigma_c = \frac{\sum \text{Real } P - \sum \text{Real } Z}{P - Z}$$

$$= \frac{(-12) - (-1.33)}{2}$$

$$= -5.33$$

⇒ Step 5 - Break Away points.

→ Characteristics eqⁿ

$$\Rightarrow 1 + G(s)H(s) = 0,$$

$$\Rightarrow 1 + \frac{K(s+4/3)}{s^3(s+12)} = 0$$

$$\Rightarrow K = - \frac{s^3 + 12s^2}{s + 4/3}$$

$$- \frac{dK}{ds} = 0 \Rightarrow \frac{(s + 4/3)(3s^2 + 24s) - (s^3 + 12s^2)(1)}{(s + 4/3)^2} = 0$$

$$\Rightarrow 3s^3 + 24s^2 + 4s^2 + 32s - s^3 - 12s^2 = 0$$

$$\Rightarrow 2s^3 + 16s^2 + 32s = 0$$

$$\Rightarrow \boxed{s=0}, \Rightarrow s^2 + 8s + 16 = 0$$

$$\Rightarrow (s+4)^2 = 0$$

$$\Rightarrow \boxed{s=-4}$$

4. Obtain Root-Locus Plot for the Unity Feedback System with System Gain:

$$\Rightarrow G(S) = \frac{K}{S(S+1)(S+2)(S+3)}$$

⇒ Step-1 - Obtain Loci

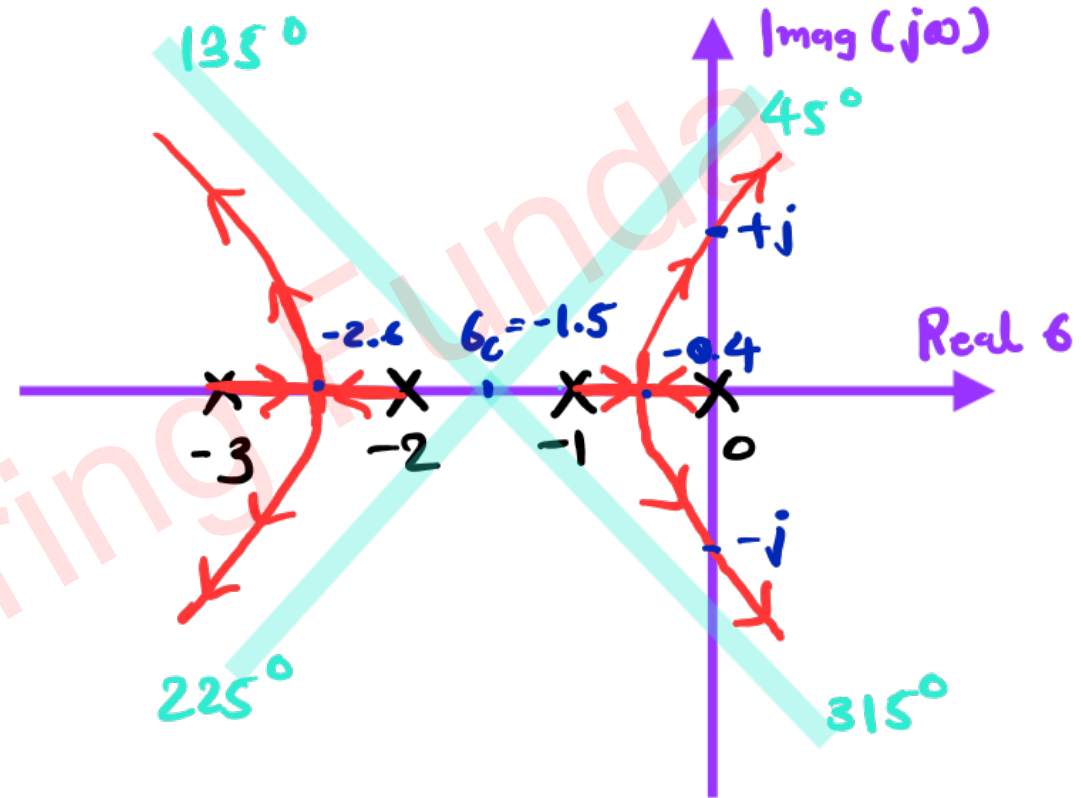
- $P_1 = 0, P_2 = -1, P_3 = -2, P_4 = -3$
- Total poles $P = 4$
- Total zeros $Z = 0$
- Total loci = $\max(P, Z)$
 $= 4$

⇒ Step-2 - Total Asymptotes.

$$- X = P - Z = 4 - 0 = 4$$

⇒ Step-3 - Angle of Asymptotes.

$$- \theta = \frac{(2n+1)}{P-Z} 180^\circ, \quad n = 0, 1, 2, 3 = 45^\circ, 135^\circ, 225^\circ, 315^\circ$$



→ Step 4 - Centroid of Asymptotes.

$$\sigma_c = \frac{\sum \text{Real } p - \sum \text{Real } z}{p - z}$$

$$= \frac{(0 - 1 - 2 - 3) - 0}{4 - 0} = -1.5$$

→ Step 5 - Break Away point

- Characteristic eq.ⁿ

$$\Rightarrow 1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{K}{s(s+1)(s+2)(s+3)} = 0$$

$$\Rightarrow K = -s(s+1)(s+2)(s+3)$$

$$\Rightarrow K = -s(s^2 + 3s + 2)(s+3)$$

$$\Rightarrow K = -s(s^3 + 3s^2 + 3s^2 + 9s + 2s + 6)$$

$$\Rightarrow K = -(s^4 + 6s^3 + 11s^2 + 6s)$$

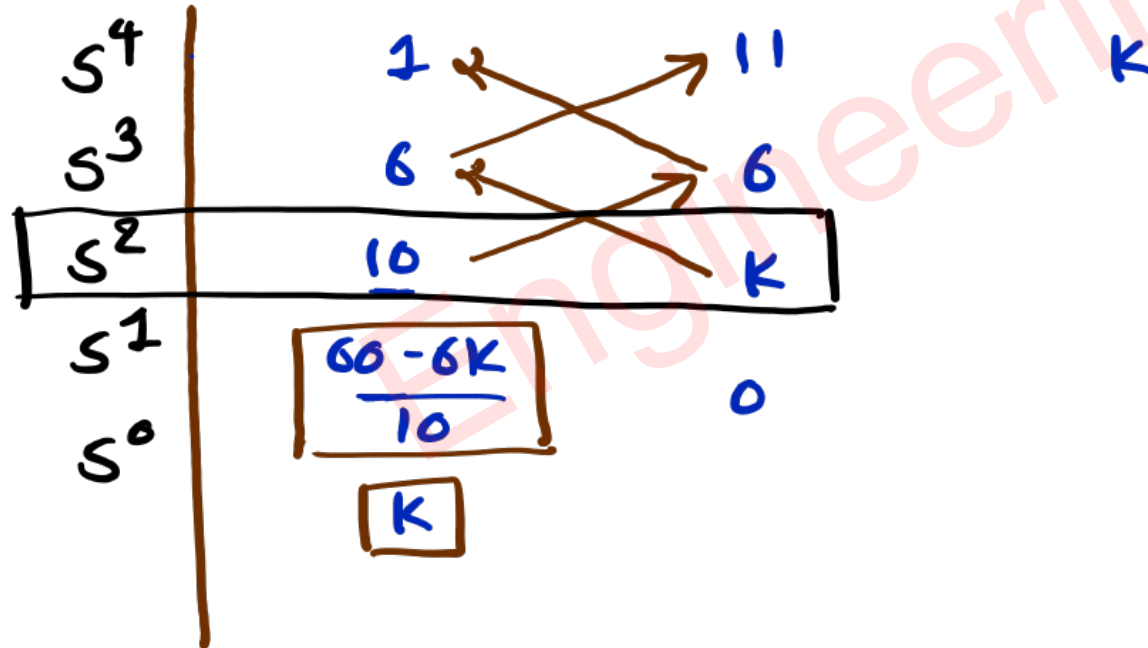
$$- \frac{dk}{ds} = 0 \Rightarrow 4s^3 + 18s^2 + 22s + 6 = 0$$

$$\Rightarrow s = -0.4, s = -2.6$$

- Step-6 - Intersection to Imag Axis.

- Characteristics eq.ⁿ

$$\Rightarrow s^4 + 6s^3 + 11s^2 + 6s + K = 0$$



$$- \frac{60 - 6K}{10} = 0 \Rightarrow 6K = 60 \\ \Rightarrow \boxed{K = 10}$$

- Auxillary eqⁿ

$$\Rightarrow 10s^2 + K = 0$$

$$\Rightarrow 10s^2 + 10 = 0$$

$$\Rightarrow s^2 = -1$$

$$\Rightarrow s = \pm j$$

Engineering Funda

5. Obtain Root-Locus Plot for the Unity Feedback System with System Gain:

$$\Rightarrow G(S) = \frac{K(S + 2)}{(S + 1)(S + 3 + j2)(S + 3 - j2)}$$

⇒ Step 1 - Obtain Loci

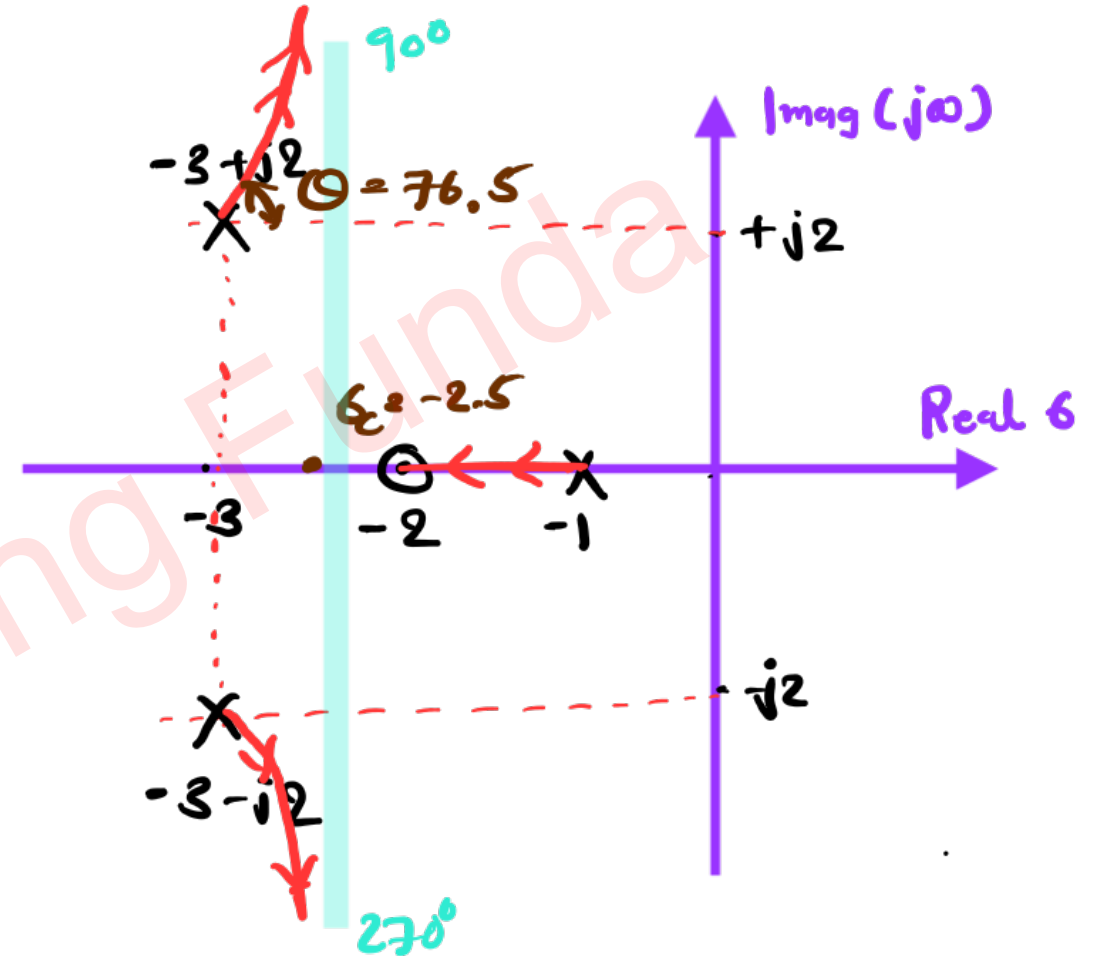
- $Z_1 = -2$
- $P_1 = -1, P_2 = -3 - j2, P_3 = -3 + j2$
- Total poles $p = 3$
- Total zeros $z = 1$
- Total loci = $\max(P, Z) = 3$

⇒ Step 2 - Total Asymptotes

$$\begin{aligned} - X &= P - Z \\ &= 3 - 1 \\ &= 2 \end{aligned}$$

- Step 6 Angle of Departure

$$\begin{aligned} \Rightarrow \theta &= 180^\circ - (\sum \theta_p - \sum \theta_z) \\ &= 180^\circ - (90 + 135 - 116.5) \end{aligned}$$



$$= 76.5$$

⇒ Step-3 - Angle of Asymptotes

$$\begin{aligned} - \theta &= \frac{(2n+1)}{P-2} 180^\circ, \quad n=0, 1 \\ &= 90^\circ, 270^\circ \end{aligned}$$

⇒ Step-4 - Centroid of Asymptotes.

$$\begin{aligned} - \sigma_c &= \frac{\sum \text{Real } p - \sum \text{Real } z}{P-2} \\ &= \frac{(-1-3-3) - (-2)}{2} \\ &= -5/2 \\ &= -2.5 \end{aligned}$$

6. Find Root Locus for given Characteristics Equation:

$$\Rightarrow s^2 + 2s + 2 + K(s + 2) = 0$$

⇒ Characteristics eq.ⁿ

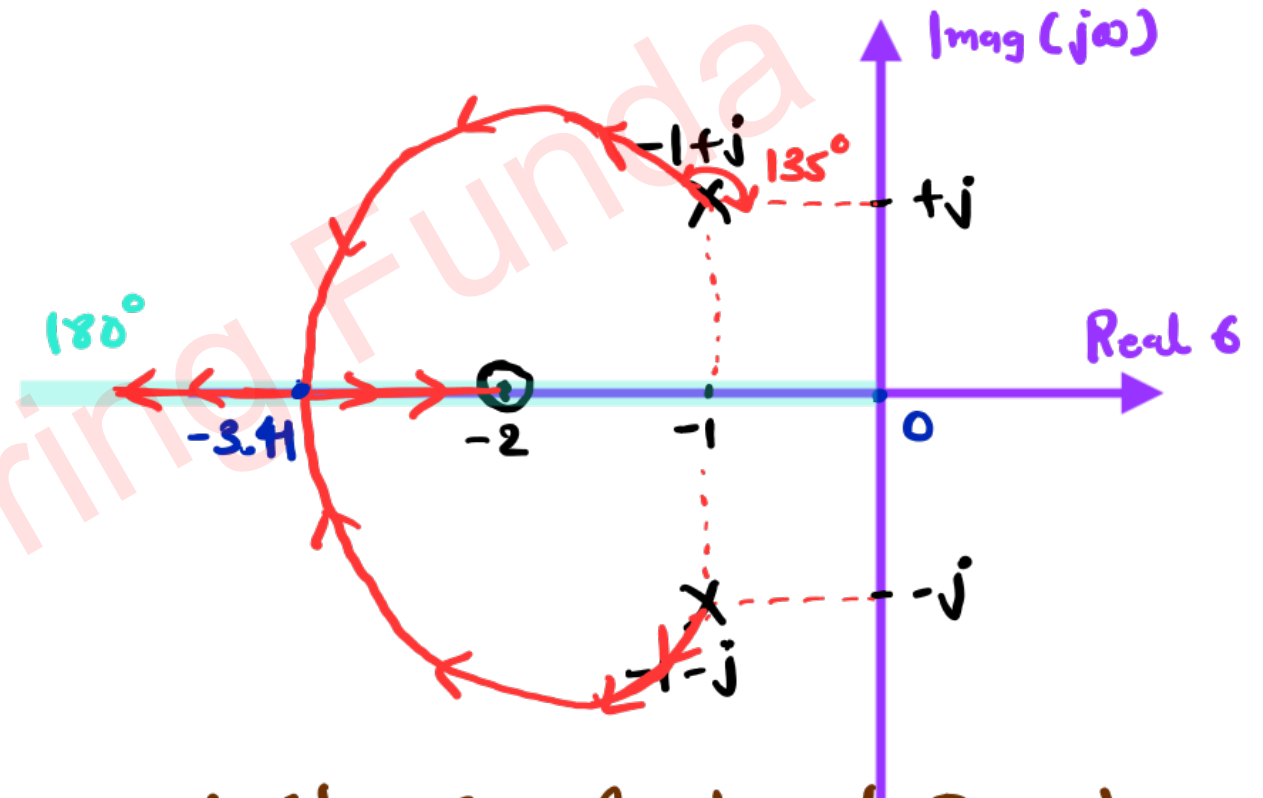
$$\Rightarrow 1 + G(s)H(s) = 0$$

$$\Rightarrow s^2 + 2s + 2 + K(s + 2) = 0$$

$$\Rightarrow 1 + \frac{K(s+2)}{s^2 + 2s + 2} = 0$$

⇒ Open loop system

$$\begin{aligned} \Rightarrow G(s)H(s) &= \frac{K(s+2)}{s^2 + 2s + 2} \\ &= \frac{K(s+2)}{(s+1)^2 - j^2} \\ &= \frac{K(s+2)}{(s+1+j)(s+1-j)} \end{aligned}$$



⇒ Step-5 - Angle of Departure.

$$\begin{aligned} \Rightarrow \theta_d &= 180^\circ - (\angle O_1 - \angle O_2) \\ &= 180^\circ - (90^\circ - 45^\circ) \\ &= 135^\circ \end{aligned}$$

~> Step-1 - Obtain loci

- $z_1 = -2$

- $p_1 = -1-j$, $p_2 = -1+j$

- Total poles $P = 2$

- Total zeros $Z = 1$

- Total loci $= \max(P, Z)$
 $= 2$

~> Step-2 - Total Asymptotes.

- $X = P - Z = 2 - 1 = 1$

~> Step-3 - Angle of Asymptote

- $\theta = \frac{(2n+1)}{P-Z} 180^\circ$, $n = 0$

$= 180^\circ$

~) Step-4 - Centroid of Asymptote.

$$- \sigma_c = \frac{\sum \text{Real } p - \sum \text{Real } z}{P - Z}$$

$$= \frac{(-1-1) - (-2)}{1} = 0$$

~) Step-6 - Break Away point.

- Characteristic eqⁿ

$$\Rightarrow s^2 + 2s + 2 + K(s+2) = 0$$

$$\Rightarrow K = - \frac{s^2 + 2s + 2}{s+2}$$

$$- \frac{dK}{ds} = 0 \Rightarrow \frac{(s+2)(2s+2) - (s^2 + 2s + 2)(1)}{(s+2)^2} = 0$$

$$\Rightarrow 2s^2 + \cancel{2s} + 4s + 4 - s^2 - \cancel{2s} - 2 = 0$$

$$\Rightarrow s^2 + 4s + 2 = 0$$

$$\Rightarrow s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$s = \frac{-4 \pm \sqrt{4^2 - 4 \times 2}}{2}$$

$$s = \frac{-3.41}{\checkmark}, \frac{-0.58}{\times}$$

7. Obtain Root-Locus Plot for the Unity Feedback System with System Gain:

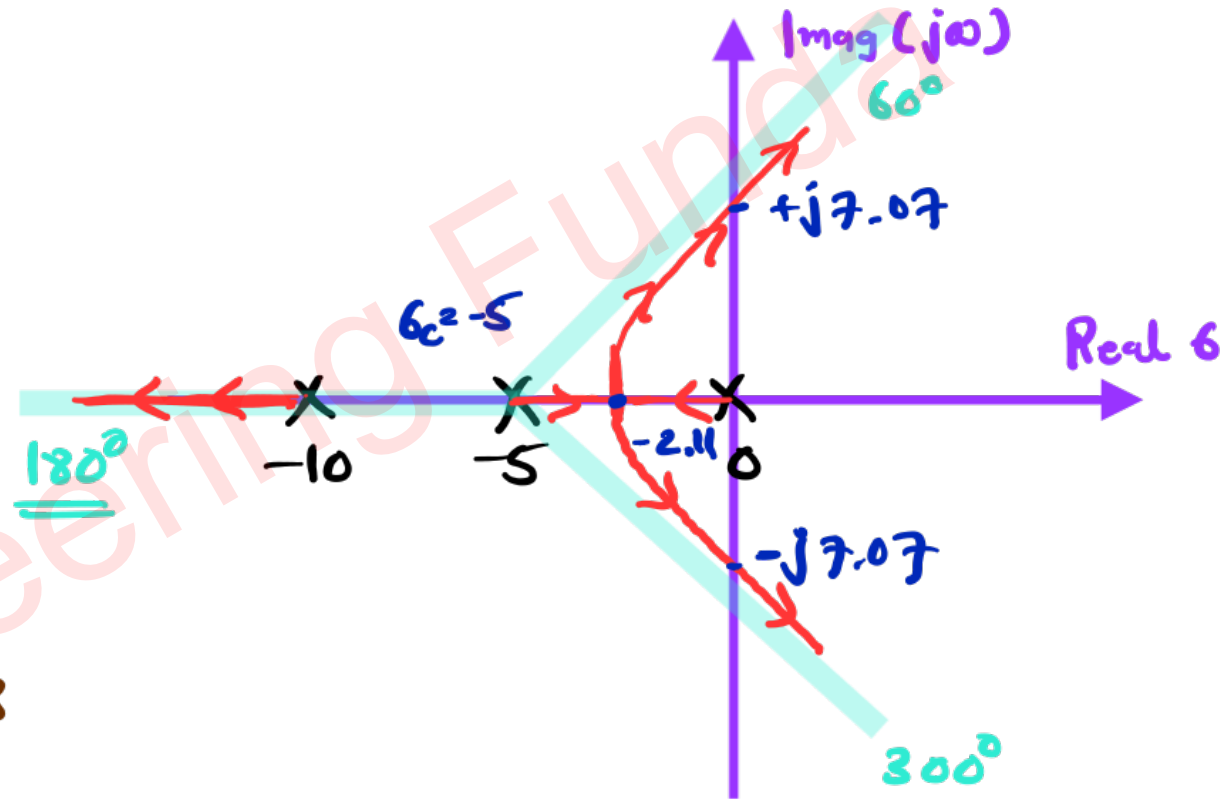
$$\Rightarrow G(S) = \frac{K}{S(S+5)(S+10)}$$

⇒ Step-1 - Obtain loci

- $P_1 = 0, P_2 = -5, P_3 = -10$
- Total poles $P = 3$
- Total zeros $Z = 0$
- Total loci = $\max(P, Z)$
 $= 3$

⇒ Step-2 - Total Asymptotes

$$\begin{aligned} X &= P - Z \\ &= 3 - 0 \\ &= 3 \end{aligned}$$



~) Step 3 - Angles of Asymptotes.

$$\begin{aligned} - \theta &= \frac{(2n+1)}{P-Z} 180^\circ, \quad n = 0, 1, 2 \\ &= 60^\circ, 180^\circ, 300^\circ \end{aligned}$$

~) Step 4 - Centroid of Asymptotes.

$$\begin{aligned} - \sigma_c &= \frac{\sum \text{Real } P - \sum \text{Real } Z}{P-Z} \\ &= \frac{(0-5-10) - 0}{3} \\ &= -5 \end{aligned}$$

~) Step 5 - Break Away point

- Characteristic eqⁿ

$$\Rightarrow 1 + K(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{K}{s(s+5)(s+10)} = 0$$

$$\Rightarrow K = -s(s+5)(s+10)$$

$$\Rightarrow K = -(s^3 + 15s^2 + 50s)$$

$$- \frac{dK}{ds} = 0 \Rightarrow 3s^2 + 30s + 50 = 0$$

$$\Rightarrow s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-30 \pm \sqrt{900 - 4(3)(50)}}{6}$$

$$= \frac{-7.88}{\times}, \frac{-2.11}{\checkmark}$$

~ Step-6 - Intersection to Imag. Axis

- Characteristic eq.ⁿ

$$\Rightarrow s^3 + 15s^2 + 50s + K = 0$$

s^3	+	1	$\swarrow$	$\searrow$	50
s^2	+	15	$\swarrow$	$\searrow$	K
s^1	+	$\frac{750-K}{15}$			0
s^0	+	K			

- For marginal stable system

$$\Rightarrow \frac{750-K}{15} = 0 \Rightarrow \boxed{K = 750}$$

~ Auxillary eq.ⁿ

$$\Rightarrow 15s^2 + K = 0$$

$$\Rightarrow 15s^2 + 750 = 0$$

$$\Rightarrow s^2 = -50$$

$$\Rightarrow s = \pm j 7.07$$

8. Obtain Root-Locus Plot for given open Loop transfer function of the System:

$$\Rightarrow G(S)H(S) = \frac{K}{S(S+2+j2)(S+2-j2)}$$

~> Step-1 - Obtain Loci

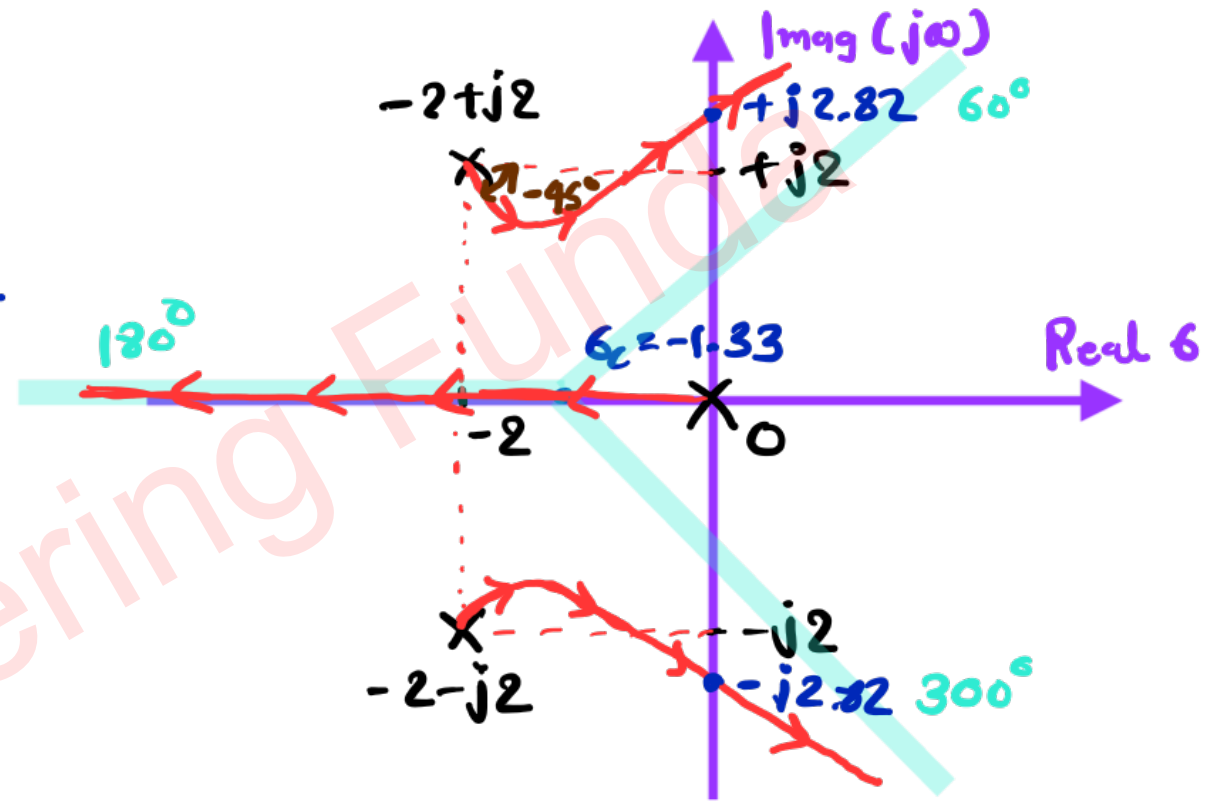
- $P_1 = 0, P_2 = -2-j2, P_3 = -2+j2$
- Total poles $P = 3$
- Total zeros $Z = 0$
- Total loci = $\max(P, Z)$
 $= 3$

~> Step-2 - Total Asymptotes

$$- X = P - Z = 3 - 0 = 3$$

~> Step-3 - Angle of Asymptotes

$$- \theta = \frac{(2n+1)}{P-Z} 180^\circ, \quad n = 0, 1, 2 = 60^\circ, 180^\circ, 300^\circ$$



~> Step-5 - Angle of Departure.

$$\begin{aligned}\Rightarrow \theta_d &= 180^\circ - (\angle \theta_p - \angle \theta_z) \\ &= 180^\circ - (135^\circ + 90^\circ) \\ &= -45^\circ\end{aligned}$$

~) Step 4 - Centroid of Asymptotes

$$\sigma_c = \frac{\sum \text{Real } p - \sum \text{Real } z}{P - Z}$$

$$= \frac{(0 - 2 - 2) - 0}{3 - 0}$$

$$= -4/3 = -1.33$$

~) Step-6 - Intersection to Imag. Axis.

→ Characteristic eq.ⁿ

$$\Rightarrow 1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{K}{s(s+2+j2)(s+2-j2)} = 0$$

$$\Rightarrow s((s+2)^2 - j^2 4) + K = 0$$

$$\Rightarrow s(s^2 + 4s + 4 + 4) + K = 0$$

$$\Rightarrow s^3 + 4s^2 + 8s + K = 0$$

- Routh matrix

s^3	+	8	
s^2	+	4	K
s^1	+	$\frac{32-K}{4}$	0
s^0	+	K	

- Marginal stable cond.ⁿ

$$\Rightarrow \frac{32-K}{4} = 0$$

$$\Rightarrow \boxed{K = 32}$$

- Auxillary eqⁿ

$$\Rightarrow 4s^2 + K = 0$$

$$\Rightarrow 4s^2 + 32 = 0$$

$$\Rightarrow s^2 = -8$$

$$\Rightarrow s = \pm j 2.82$$

9. Obtain Root-Locus Plot for given open Loop transfer function of the System:

$$\Rightarrow G(S)H(S) = \frac{K(S+2)}{(S+1+j\sqrt{3})(S+1-j\sqrt{3})}$$

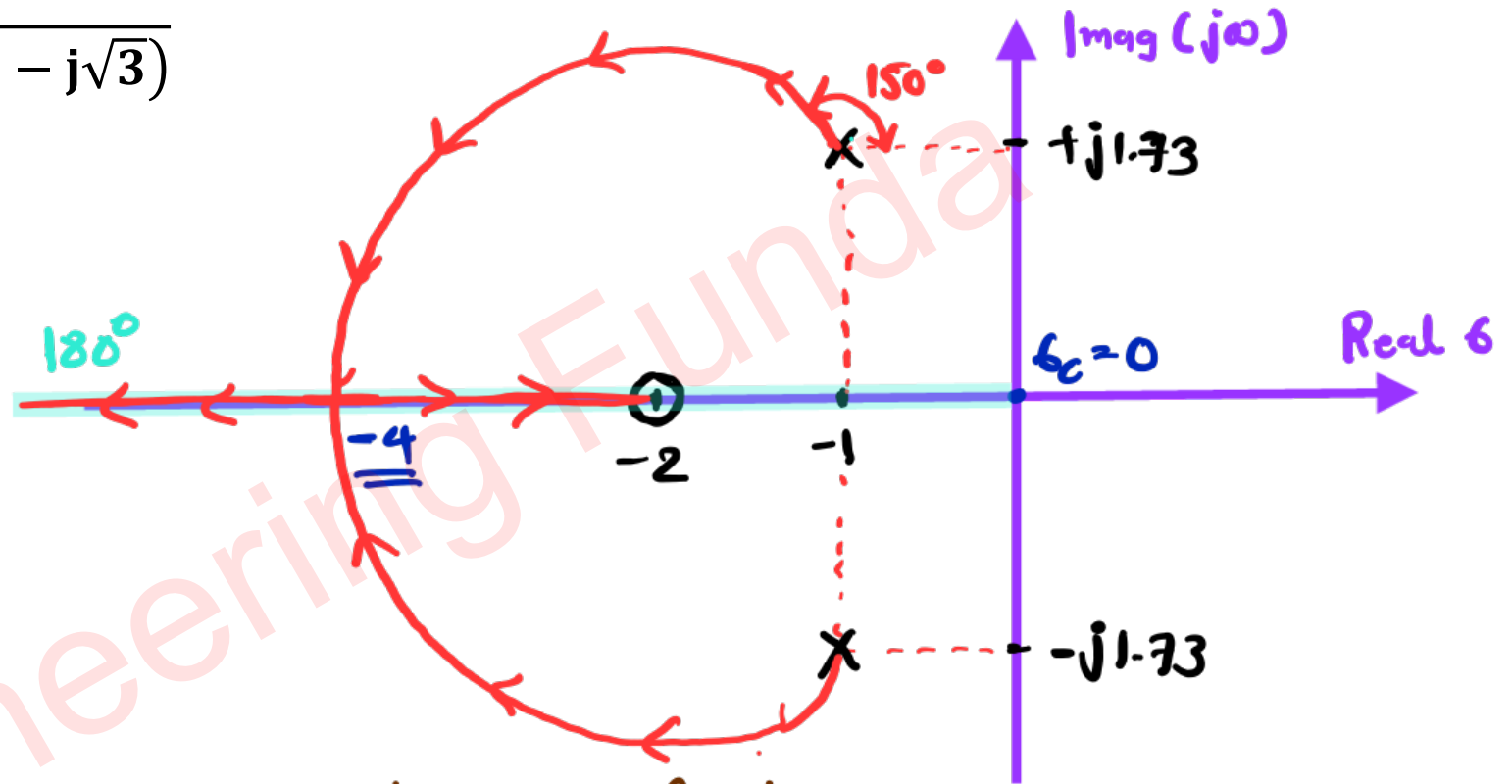
⇒ Step-1 - Obtain loci

- $Z_1 = -2$
- $P_1 = -1-j\sqrt{3}, P_2 = -1+j\sqrt{3}$
- Total poles $P = 2$
- Total zeros $Z = 1$
- Total loci = $\max(P, Z)$
 $= 2$

⇒ Step-2 - Total Asymptotes.

$$- X = P - Z = 2 - 1 = 1$$

⇒ Step-3 - Angle of Asymptote.



⇒ Step-5 - Angle of Departure

$$\begin{aligned} - \theta_d &= 180^\circ - (\sum \theta_1 - \sum \theta_2) \\ &= 180^\circ - (90^\circ - 60^\circ) \\ &= 150^\circ \end{aligned}$$

$$- \theta = \frac{(2n+1)}{p-2} 180^\circ, \quad n=0$$

$$= 180^\circ$$

→ Step 4 - Centroid of Asymptote

$$- \sigma_c = \frac{\sum \text{Real } p - \sum \text{Real } z}{p-2}$$

$$= \frac{(-1-1) - (-2)}{2-1} = 0$$

→ Step 6 - Break Away point

- Characteristic eqⁿ

$$\Rightarrow 1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{K(s+2)}{(s+1+j\sqrt{3})(s+1-j\sqrt{3})} = 0$$

$$\Rightarrow K = - \frac{(s+1+j\sqrt{3})(s+1-j\sqrt{3})}{(s+2)}$$

$$\Rightarrow K = - \frac{((s+1)^2 - j^2 3)}{(s+2)}$$

$$\Rightarrow K = - \frac{(s^2 + 2s + 4)}{(s+2)}$$

$$- \frac{dK}{ds} = 0 \Rightarrow \frac{(s+2)(2s+2) - (s^2 + 2s + 4)(1)}{(s+2)^2} = 0$$

$$\Rightarrow 2s^2 + 6s + \cancel{4} - s^2 - 2s - \cancel{4} = 0$$

$$\Rightarrow s^2 + 4s = 0$$

$$\Rightarrow \frac{s=0}{\times}, \frac{s=-4}{\checkmark}$$

10. Obtain Root-Locus Plot for given open Loop transfer function of the System:

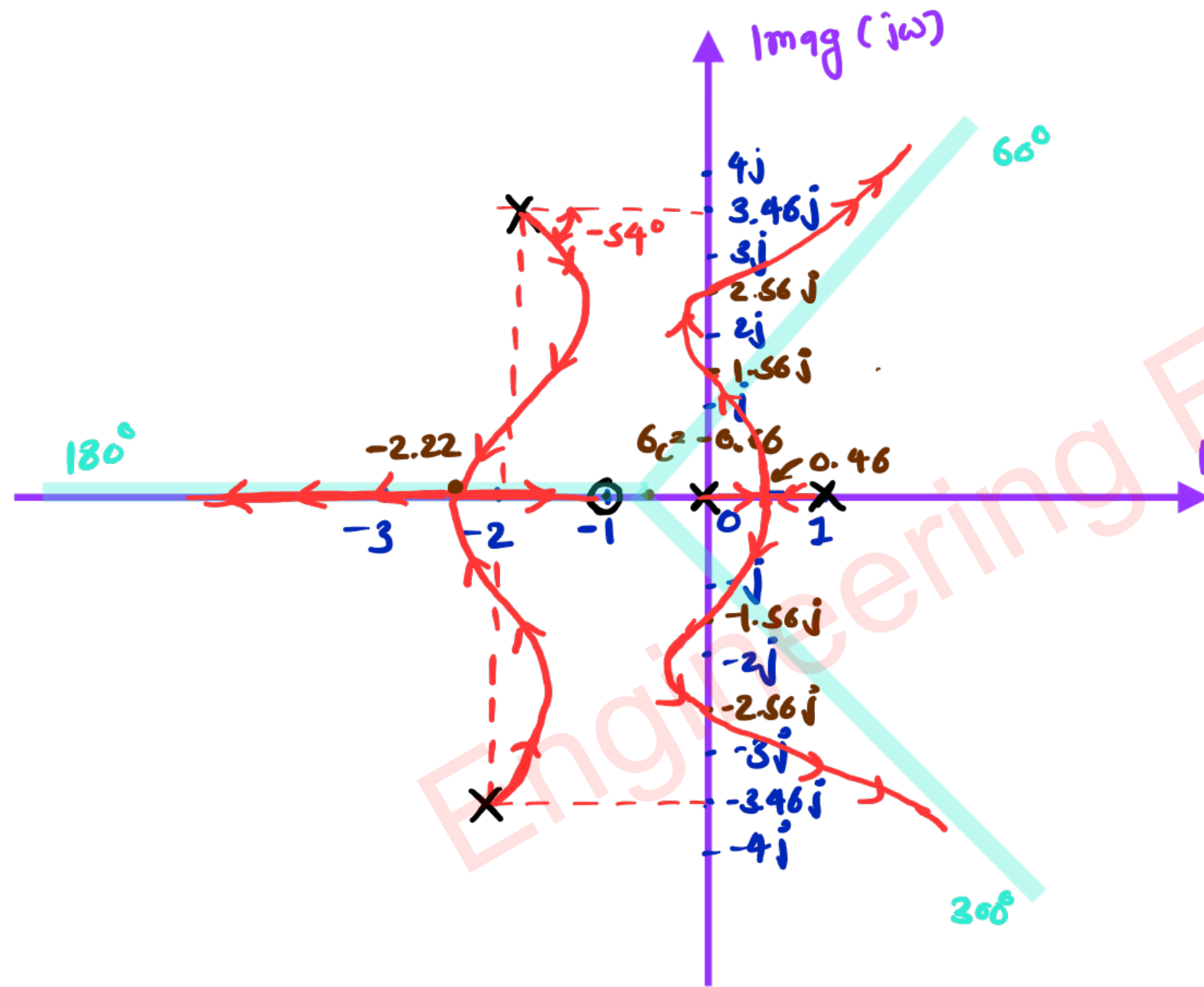
$$\Rightarrow G(S)H(S) = \frac{K(S+1)}{S(S-1)(S^2+4S+16)}$$

$$\Rightarrow G(S)H(S) = \frac{K(S+1)}{S(S-1)((S+2)^2 - j^2 12)}$$

$$\Rightarrow G(S)H(S) = \frac{K(S+1)}{S(S-1)(S+2+j3.46)(S+2-j3.46)}$$

⇒ Step-1 - Obtain loci

- $z_1 = -1$
- $p_1 = 0, p_2 = 1, p_3 = -2 - j3.46, p_4 = -2 + j3.46$
- Total poles $P = 4$
- Total zeros $Z = 1$
- Total loci = $\max(P, Z) = 4$



⇒ Step-5 - Angle of Departure

$$\begin{aligned}
 -\theta_d &= 180^\circ - (\angle\phi_p - \angle\phi_z) \\
 &= 180^\circ - (90^\circ + 120^\circ + 130^\circ - 106^\circ) \\
 &= -54^\circ
 \end{aligned}$$

~> Step-2 - Total Asymptotes.

$$- X = P - Z = 4 - 1 = 3$$

~> Step-3 - Angle of Asymptotes

$$\begin{aligned} - \theta &= \frac{(2n+1)}{P-Z} 180^\circ, \quad n = 0, 1, 2 \\ &= 60^\circ, 180^\circ, 300^\circ \end{aligned}$$

~> Step-4 - Centroid of Asymptotes.

$$- \sigma_c = \frac{\sum \text{Real } P - \sum \text{Real } Z}{P-Z}$$

$$= \frac{(0+1-2-2) - (-1)}{4-1}$$

$$= -2/3$$

$$= -0.66$$

Step-6 - Break Away point

- Characteristic eq.ⁿ

$$\Rightarrow 1 + K(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{K(s+1)}{s(s-1)(s^2+4s+16)} = 0$$

$$\Rightarrow K = - \frac{s(s-1)(s^2+4s+16)}{(s+1)}$$

$$\Rightarrow K = - \frac{s(s^3+4s^2+16s-s^2-4s-16)}{(s+1)}$$

$$\Rightarrow K = - \left(\frac{s^4 + 3s^3 + 12s^2 - 16s}{s+1} \right)$$

$$- \frac{dK}{ds} = 0 \Rightarrow \frac{(s+1)(4s^3 + 9s^2 + 24s - 16) - (s^4 + 3s^3 + 12s^2 - 16s)(1)}{(s+1)^2} = 0$$

$$\Rightarrow \underline{4s^4} + \underline{9s^3} + \underline{24s^2} - \underline{16s} + \underline{4s^3} + \underline{9s^2} + \underline{24s} - 16 - \underline{s^4} - \underline{3s^3} - \underline{12s^2} + \underline{16s} = 0$$

$$\Rightarrow 3s^4 + 10s^3 + 21s^2 + 24s - 16 = 0$$

$$\Rightarrow s = -2.22, s = 0.46$$

~) Step-7 - Intersection to Imag. Axis

- characteristic eq.ⁿ

$$\Rightarrow s^4 + 3s^3 + 12s^2 - 16s + K(s+1) = 0$$

$$\Rightarrow s^4 + 3s^3 + 12s^2 + s(K-16) + K = 0$$

Routh matrix

s^4	1	12
s^3	3	$K-16$
s^2	$\frac{52-K}{3}$	K

$$= \frac{3(12) - (K-16)}{3}$$

$$= \frac{52-K}{3}$$

$$\begin{array}{c|c} s^1 & \frac{-k^2 + 59k - 832}{52 - k} \\ s^0 & k \end{array}$$

- marginal stability

$$\Rightarrow \frac{-k^2 + 59k - 832}{52 - k} = 0$$

$$\Rightarrow -k^2 + 59k - 832 = 0$$

$$\Rightarrow k = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\Rightarrow k = 35.68, \quad k = 23.31$$

0

$$= \frac{\left(\frac{52-k}{3}\right)(k-16) - 3k}{\left(\frac{52-k}{3}\right)}$$

$$= \frac{-k^2 + 68k - 832 - 9k}{52 - k}$$

$$= \frac{-k^2 + 59k - 832}{52 - k}$$

- Auxillary eq.ⁿ

$$\Rightarrow \left(\frac{s^2 - K}{3} \right) s^2 + K = 0$$

$$\Rightarrow s^2 = \frac{-3K}{s^2 - K}$$

$$\Rightarrow s = \pm j \sqrt{\frac{3K}{s^2 - K}}$$

$$\Rightarrow s = \pm j 2.56, \quad s = \pm j 1.56$$

11. Obtain Root-Locus Plot for given open Loop transfer function of the System:

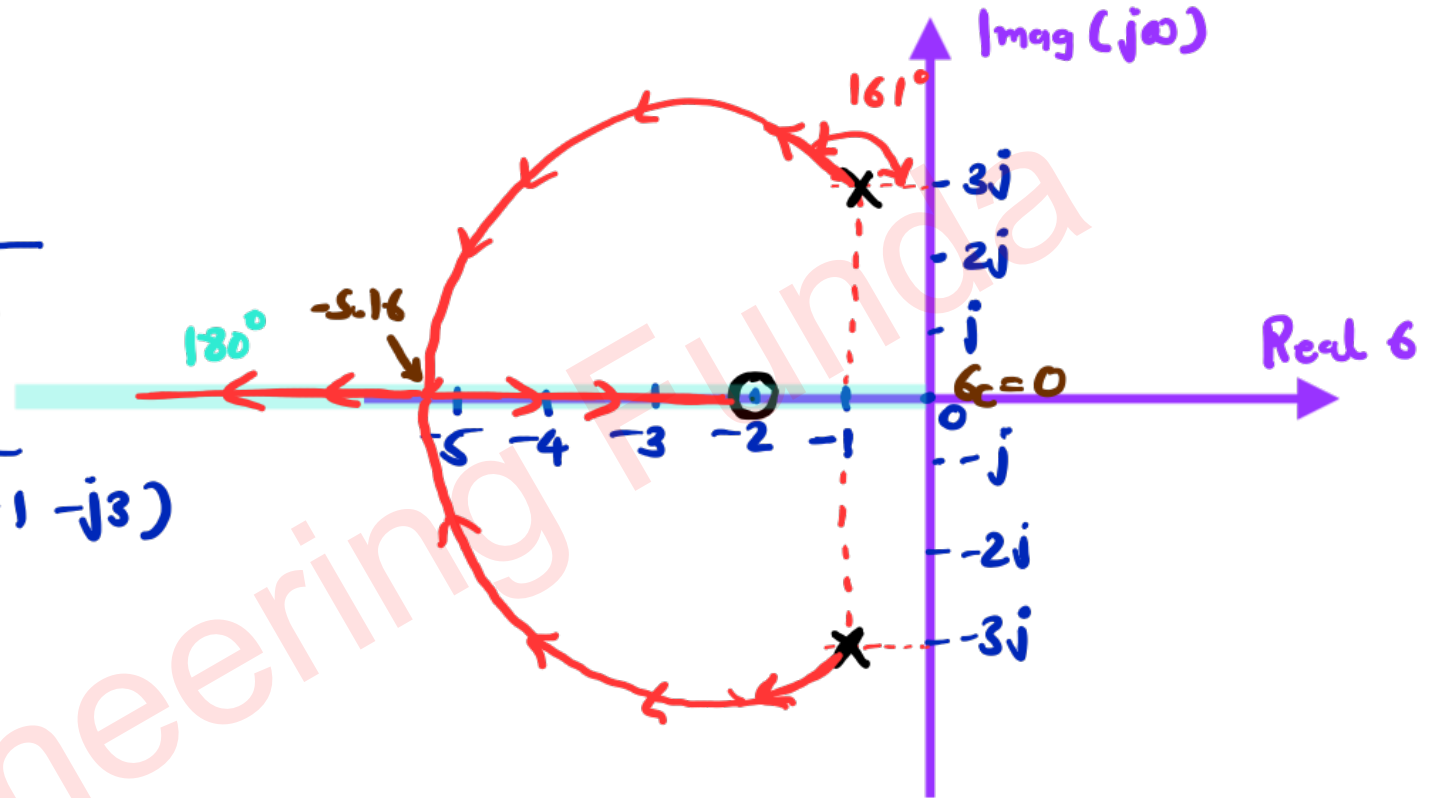
$$\Rightarrow G(S)H(S) = \frac{K(S+2)}{S^2 + 2S + 10}$$

$$\Rightarrow G(S)H(S) = \frac{k(S+2)}{(S+1)^2 - j^2 9}$$

$$\Rightarrow G(S)H(S) = \frac{k(S+2)}{(S+1+j3)(S+1-j3)}$$

∴ Step 1 - Obtain loci

- $z_1 = -2$
- $p_1 = -1-j3$, $p_2 = -1+j3$
- Total poles $p = 2$
- Total zeros $z = 1$
- Total loci = $\max(p, z) = 2$



∴ Step 5 - Angle of Departure

$$\begin{aligned}\Rightarrow \theta_d &= 180^\circ - (\sum \theta_p - \sum \theta_z) \\ &= 180^\circ - (90^\circ - 71.56^\circ) \\ &= 161^\circ\end{aligned}$$

⇒ Step-2 - Total Asymptotes

$$- X = P - 2 = 2 - 1 = 1$$

⇒ Step-3 - Angle of Asymptotes

$$\begin{aligned} - \theta &= \frac{(2n+1)}{P-2} 180^\circ, \quad n=0 \\ &= 180^\circ \end{aligned}$$

⇒ Step-4 - Centroid of Asymptotes

$$\begin{aligned} - \sigma_c &= \frac{\sum \text{Real } P - \sum \text{Real } Z}{P-2} \\ &= \frac{(-1-1) - (-2)}{2-1} \\ &= 0 \end{aligned}$$

→ Step 6 - Break Away point.

→ Characteristic eqⁿ

$$\Rightarrow 1 + K(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{K(s+2)}{s^2+2s+10} = 0$$

$$\Rightarrow K = - \left(\frac{s^2+2s+10}{s+2} \right)$$

$$\Rightarrow \frac{dK}{ds} = 0 \Rightarrow \frac{(s+2)(2s+2) - (s^2+2s+10)(1)}{(s+2)^2} = 0$$

$$\Rightarrow 2s^2 + 6s + 4 - s^2 - 2s - 10 = 0$$

$$\Rightarrow s^2 + 4s - 6 = 0$$

$$\Rightarrow s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-4 \pm \sqrt{4^2 + 4 \times 6}}{2} = \frac{1.16}{\times} \& \frac{-5.16}{\vee}$$

Examples of Root Locus

1. **Example:** Consider a closed Loop system shown in Figure (a) below, The root locus for it is shown in figure (b). The closed-loop transfer function for the system is

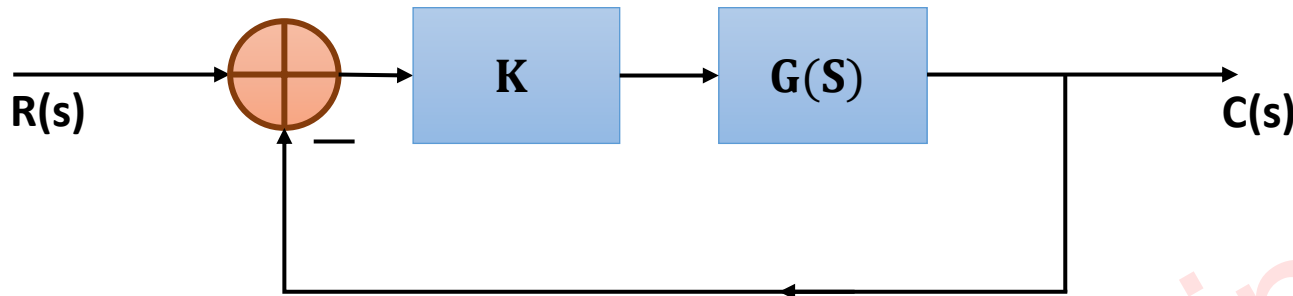


Figure (a)

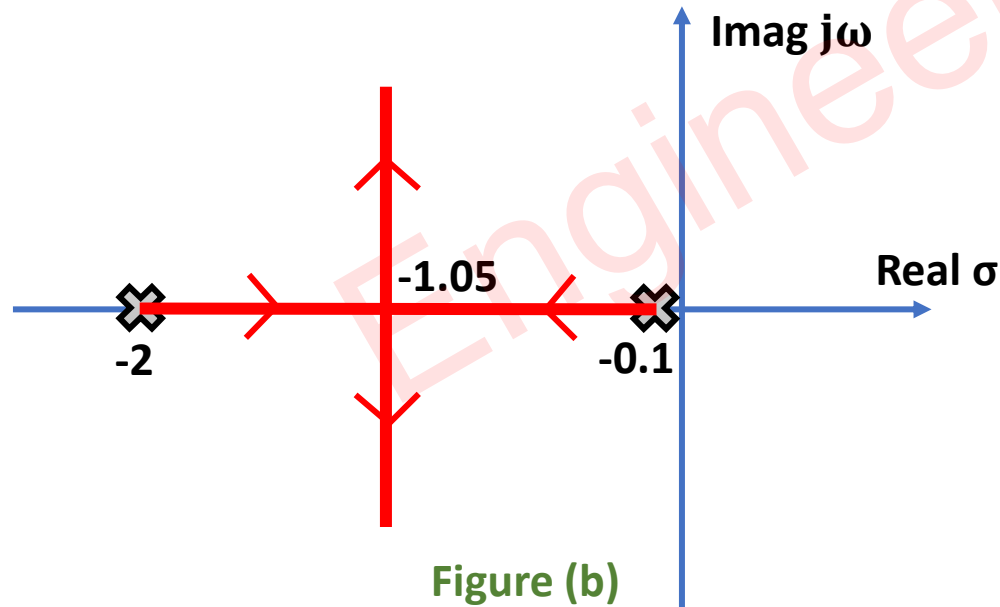


Figure (b)

$$\sim G(s) = \frac{1}{(s+0.1)(s+2)}$$

$$\sim H(s) = 1$$

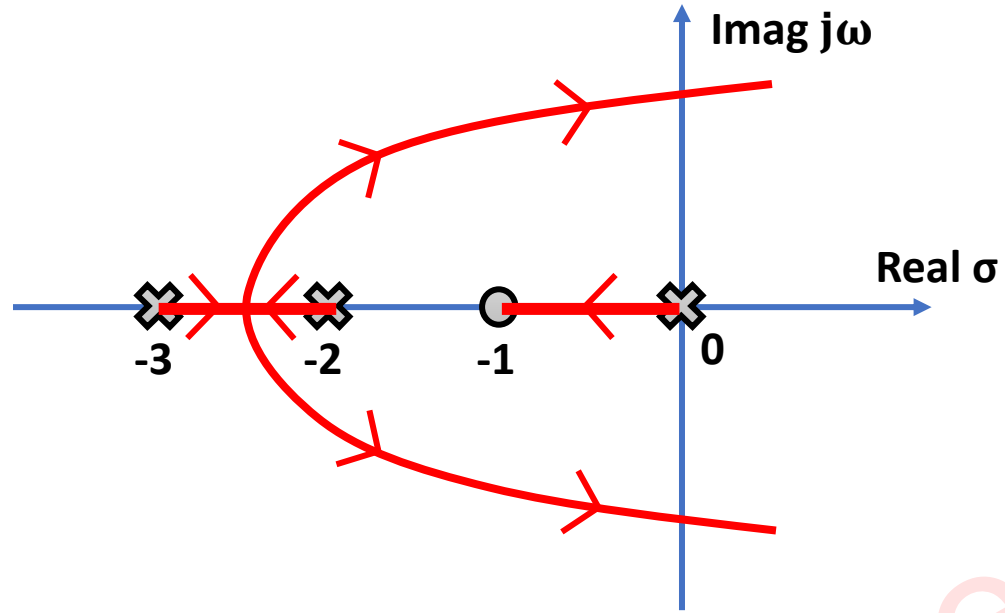
$\sim$ closed loop T.F.

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{KG(s)}{1+KG(s)}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{K/(s+0.1)(s+2)}{1+K/(s+0.1)(s+2)}$$

$$\Rightarrow \frac{C(s)}{R(s)} = \frac{K}{K+(s+0.1)(s+2)}$$

2. **Example:** The Root Locus Plot for a system is given below. The Open Loop Transfer Function corresponding to this plot is given by:



→ Open loop T. F. :

$$\Rightarrow G(s)H(s) = \frac{K(s+1)}{s(s+2)(s+3)}$$

3. **Example:** Given $G(S)H(S) = \frac{K}{s(s+1)(s+3)}$, The point of interaction of asymptotes of the root loci with the real axis is

$$\Rightarrow \sigma_c = \frac{\sum \text{Real } P - \sum \text{Real } Z}{P - Z}$$

$$= \frac{(0 - 1 - 3) - 0}{3 - 0}$$

$$= -4/3 = -1.33$$

- Total zeros $Z = 0$

- Total poles $P = 3$

- $P_1 = 0, P_2 = -1, P_3 = -3$

$\Rightarrow$ Intersection to Imag. Axis.

- Characteristic eq.ⁿ

$$\Rightarrow 1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{K}{s(s+1)(s+3)} = 0$$

$$\Rightarrow s^3 + 4s^2 + 3s + K = 0$$

- Routh matrix

s^3	1	3
s^2	4	K
s^1	$\frac{12-K}{4}$	0
s^0	K	

- marginal stability.

$$\Rightarrow \frac{12-K}{4} = 0$$

$$\Rightarrow K = 12$$

- Auxillary eq.ⁿ

$$\Rightarrow 4s^2 + K = 0$$

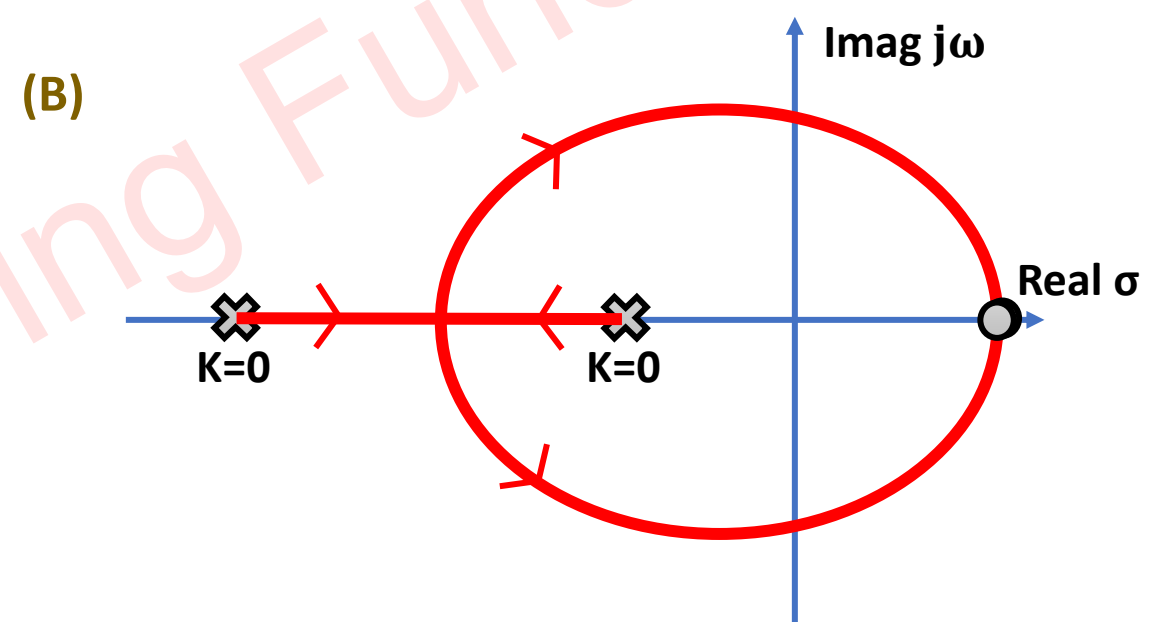
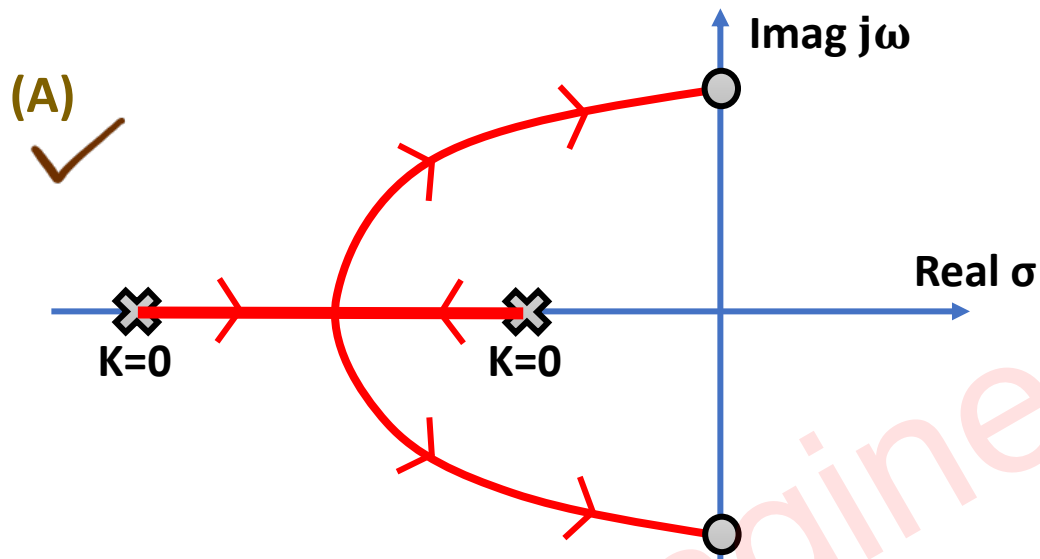
$$\Rightarrow 4s^2 + 12 = 0$$

$$\Rightarrow s^2 = -3$$

$$\Rightarrow s = \pm j\sqrt{3}$$

Examples of Root Locus

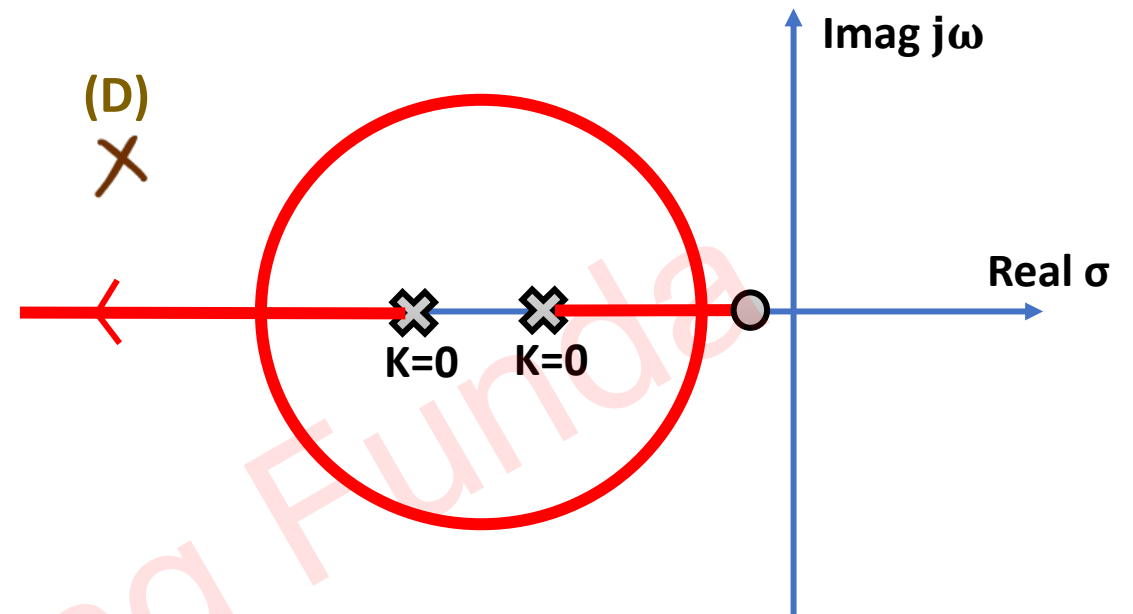
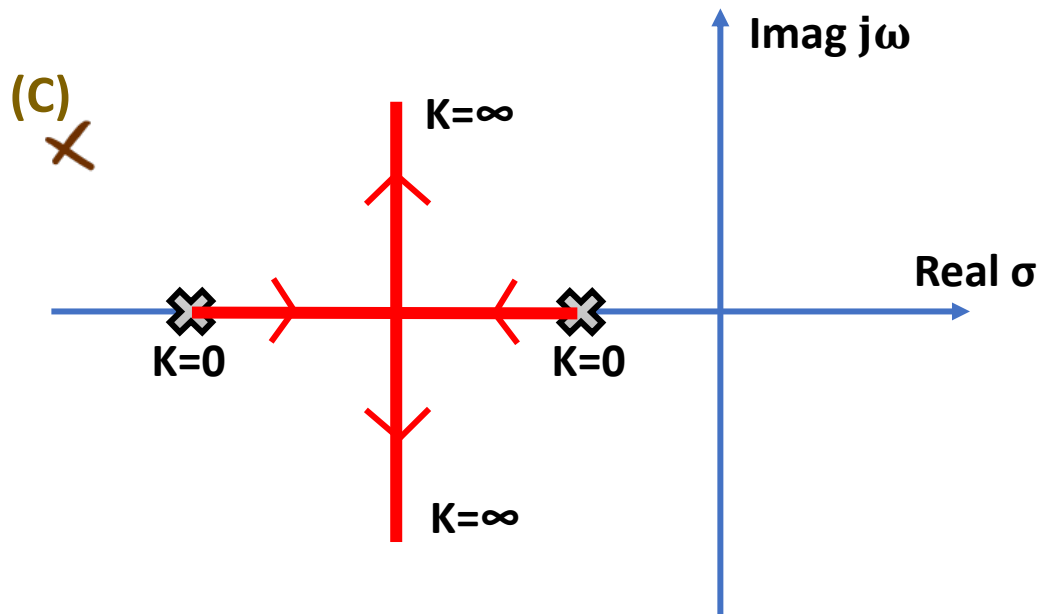
4. **Example:** The transfer function of a closed loop system is $T(S) = \frac{K(S^2+9)}{(S+1)(S+3)+K(S^2+9)}$, where k is the forward path gain. The Root Locus of system is



~ Closed Loop Transfer Function $T(s)$

$$\Rightarrow T(s) = \frac{G(s)}{1 + G(s)H(s)} = \frac{G(s)}{1 + G(s)} = \frac{K(s^2+9)}{(s+1)(s+3) + K(s^2+9)} = \frac{K(s^2+9)/(s+1)(s+3)}{1 + K(s^2+9)/(s+1)(s+3)}$$

- $H(s) = 1$



⇒ Open loop Transfer Function

$$\Rightarrow K(s) = \frac{K(s^2 + 9)}{(s+1)(s+3)}$$

$$- z_1 = -j3, \quad z_2 = +j3$$

$$- p_1 = -1, \quad p_2 = -3$$

5. **Example:** The characteristics equation of a feedback control system is given by:

$$S^3 + 5S^2 + (K + 6)S + K = 0$$

Where, $K > 0$ is a scalar variable parameter in the root locus diagram of the system. The asymptotes of the root loci for large values of K meet at a point in the S -Plane. Whose coordinates are

(Real , Imag)

A. (-3, 0)

✓ B. (-2, 0)

C. (-1, 0)

D. (2, 0)

∴ Characteristics eqⁿ ($1 + G(s)H(s) = 0$)

$$\Rightarrow S^3 + 5S^2 + (K+6)S + K = 0$$

$$\Rightarrow S^3 + 5S^2 + 6S + K(1+S) = 0$$

$$\Rightarrow 1 + \frac{K(1+S)}{S^3 + 5S^2 + 6S} = 0$$

∴ Open loop Transfer Function $G(s)H(s)$.

$$\Rightarrow G(s)H(s) = \frac{K(S+1)}{S(S+2)(S+3)}$$

$$\left| \begin{array}{l} - Z_1 = -1 \\ - P_1 = 0, P_2 = -2, P_3 = -3 \end{array} \right.$$

2) Centroid of Asymptotes.

$$\Rightarrow \sigma_c = \frac{\sum \text{Real } p - \sum \text{Real } z}{P - Z}$$

$$= \frac{(0 - 2 - 3) - (-1)}{3 - 1}$$

$$= -4/2 = -2$$

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6. **Example:** Consider the points $S_1 = -3 + j4$ and $S_2 = -3 - j2$ in the S Plane then for a system with open loop transfer function $G(S)H(S) = \frac{K}{(S+1)^4}$.

- A. S_1 is on the root locus, but not S_2
- ✓ B. S_2 is on the root locus, but not S_1
- C. Both S_1 and S_2 are on the root locus
- D. Neither S_1 nor S_2 is on the root locus

~> Step-1 - place given point in $G(S)H(S)$

~> Step-2 - Find Angle $\angle G(S)H(S)$

~> Step-3 - Check, $\angle G(S)H(S) = \pm n(180^\circ)$

$$\begin{aligned}\rightarrow G(S_1)H(S_1) &= \frac{K}{(-3+j4+1)^4} \\ &= \frac{K}{(-2+j4)^4}\end{aligned}$$

$$\begin{aligned}\rightarrow G(S_2)H(S_2) &= \frac{K}{(-3-j2+1)^4} \\ &= \frac{K}{(-2-j2)^4}\end{aligned}$$

$$\angle G(s_1)H(s_1) = 4 \left(-\tan^{-1}\left(\frac{4}{-2}\right) \right) \\ \neq \eta(180^\circ)$$

$$\rightarrow \angle G(s_2)H(s_2) = 4 \left(-\tan^{-1}\left(\frac{-2}{-2}\right) \right) \\ = 180^\circ \\ = \eta(180^\circ)$$

$\rightarrow s_1$ is not on Root locus of $G(s)H(s)$

$\rightarrow s_2$ is on Root locus of $G(s)H(s)$

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